Sri Chaitanya ACADEMY | Infinity Learn



JEE CHAMPION TEST SERIES

JEE Main Full Syllabus Test - 6

Test Date: 16th Mar 2025

INSTRUCTIONS

- The test is of 3 hours duration.
- The Test Booklet consists of 75 questions. The maximum marks are 300.
- There are three parts in the question paper (1, 2, 3) consisting of Physics, Chemistry and Mathematics, having 25 questions in each subject and each subject having two sections.
  (I) Section-A contains 20 multiple choice questions with only one correct option.
  (II) Section-B contains 5 numerical value type questions.
- Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.
- The answer should be within 0 to 9999. If the answer is in decimal, then round off to the nearest integer value (e.g., If the answer is above 10 and less than 10.5, round off to 10; if the answer is more than or equal to 10.5 but less than 11, round off to 11).
- Use blue/black ballpoint pen only for writing particulars or marking responses on the answer sheet. Use of pencil is strictly prohibited.

MATHEMATICS

Max Marks: 100

SECTION-I (SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. Let $\vec{u} = \hat{i} - \hat{j} - 2\hat{k}$, $\vec{v} = 2\hat{i} + \hat{j} - \hat{k}$, $\vec{v}.\vec{w} = 2$ and $\vec{v} \times \vec{w} = \vec{u} + \lambda\vec{v}$.-Then $\vec{u}.\vec{w}$ is equal to :

1) 1 **2)** $\frac{3}{2}$ **3)** 2 **4)** $-\frac{2}{3}$

2. Let a tangent to the curve $y^2 = 24x$ meet the curve $xy = 2$ at the points A and B. Then the mid points of such line segments AB lie on a parabola with the

1) directrix $4x = 3$ **2)** directrix $4x = -3$

3) Length of latus rectum $\frac{3}{2}$ **4)** Length of latus rectum 2

3. Let 'N' denote the number that turns up when a fair, die is rolled. If the probability that the system of equations

$x + y + z = 1$

$2x + Ny + 2z = 2$

$3x + 3y + Nz = 3$

Has unique solution is $\frac{k}{6}$, then the sum of value of 'k' and all possible values of 'N' is

1) 18 **2)** 19 **3)** 20 **4)** 21

4. The area enclosed by the curves $y^2 + 4x = 4$ and $y - 2x = 2$ is :

1) $\frac{25}{3}$ **2)** $\frac{22}{3}$ **3)** 9 **4)** $\frac{23}{3}$

5. $\tan^{-1}\left(\frac{1+\sqrt{3}}{3+\sqrt{3}}\right) + \sec^{-1}\left(\sqrt{\frac{8+4\sqrt{3}}{6+3\sqrt{3}}}\right)$ is equal to

1) $\frac{\pi}{4}$ **2)** $\frac{\pi}{2}$ **3)** $\frac{\pi}{3}$ **4)** $\frac{\pi}{6}$

6. Let PQR be a triangle. The points A, B and C are on the sides QR, RP and PQ respectively such that $\frac{QA}{AR} = \frac{RB}{BP} = \frac{PC}{CQ} = \frac{1}{2}$. Then $\frac{\text{area}(\Delta PQR)}{\text{area}(\Delta ABC)}$ is equal to :

1) 4 **2)** 3 **3)** 2 **4)** $\frac{5}{2}$

7. The mean and variance of the marks obtained by the students in a test are 10 and 4 respectively. Later, the marks of one of the students is increased from 8 to 12. If the new mean of the marks is 10.2, then their new variance is equal to :

1) 4.04 **2)** 4.08 **3)** 3.96 **4)** 3.92

8. The vector $\vec{a} = -\hat{i} + 2\hat{j} + \hat{k}$ is rotated through a right angle, passing through the y-axis in its way and the resulting vector is $\vec{b}$. Then the projection of $3\vec{a} + \sqrt{2}\vec{b}$ on $\vec{c} = 5\hat{i} + 4\hat{j} + 3\hat{k}$ is

1) $3\sqrt{2}$ **2)** 1 **3)** $\sqrt{6}$ **4)** $2\sqrt{3}$

9. Let x = 2 be a local minima of the function $f(x) = 2x^4 - 18x^2 + 8x + 12,\ x \in (-4, 4)$. If 'M' is local maximum value of the function, 'f' in (-4, **4)**, then M =

1) $12\sqrt{6} - \frac{33}{2}$ **2)** $12\sqrt{6} - \frac{31}{2}$ **3)** $18\sqrt{6} - \frac{33}{2}$ **4)** $18\sqrt{6} - \frac{31}{2}$

10. Let $f : (0, 1) \to \mathbb{R}$ be a function defined by $f(x) = \frac{1}{1 - e^{-x}}$ and $g(x) = (f(-x) - f(x))$.

Consider two statements
Statement-I: 'g' is an increasing function in (0,1)
Statement-II: 'g' is one-one in (0,1)
Then.
1) Only Statement I is true
2) Only Statement-II is true
3) Neither Statement -I nor Statement - II is true
4) Both Statement-I and Statement-II are true

11. Let 'z' be a complex number such that $\left|\frac{z - 2i}{z + i}\right| = 2.\ z \neq -i.$ Then 'z' lies on the circle of radius 2 and centre

1) (2, 0) **2)** (0, 0) **3)** (0, 2) **4)** (0, -2)

12. Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined by $f(x) = \log_{\sqrt{m}}\left\{\sqrt{2}(\sin x - \cos x) + m - 2\right\}$, for some 'm', such that the range of 'f' is [0, 2]. Then the value of 'm' is …….

1) 5 **2)** 3 **3)** 2 **4)** 4

13. The foot of perpendicular of the point (2, 0, 5) on the line $\frac{x+1}{2} = \frac{y-1}{5} = \frac{z+1}{-1}$ is (α, β, γ).

Then which of the following is NOT correct?

1) $\frac{\alpha\beta}{\gamma} = \frac{4}{15}$ **2)** $\frac{\alpha}{\beta} = -8$ **3)** $\frac{\beta}{\gamma} = -5$ **4)** $\frac{\gamma}{\alpha} = \frac{5}{8}$

14. Three urns A, B and C contain 7 red, 5 black, 5 red, 7 black and 6 red, 6 black balls, respectively. One of the urn is selected at random and a ball is drawn from it. If the ball drawn is black, then the probability that it is drawn from urn A is :

1) $\frac{4}{17}$ **2)** $\frac{5}{18}$ **3)** $\frac{7}{18}$ **4)** $\frac{5}{16}$

15. Let $f(x)=\begin{cases} -2, & -2 \le x \le 0 \\ x-2, & 0 < x \le 2 \end{cases}$ and $h(x)=f(|x|)+|f(x)|$. Then $\int_{-2}^{2} h(x)\,dx$ is equal to :

1) 2 **2)** 4 **3)** 1 **4)** 6

16. A square is inscribed in the circle $x^2+y^2-10x-6y+30=0$. One side of this square is parallel to $y = x + 3$. If (x_1, y_1) are the vertices of the square, then $\sum(x_1^2+y_1^2)$ is equal to :

1) 148 **2)** 156 **3)** 160 **4)** 152

17. The coefficients a, b, c in the quadratic equation $ax^2+bx+c=0$ are choose from the set $\{1, 2, 3, 4, 5, 6, 7, 8\}$. The probability of this equation having repeated roots is :

1) $\frac{3}{256}$ **2)** $\frac{1}{128}$ **3)** $\frac{1}{64}$ **4)** $\frac{3}{128}$

18. Let a circle 'C' of radius 1 and closer to the origin be such that the lines passing through the point (3, 2) and parallel to the coordinate axes touch it. Then the shortest distance of the circle 'C' from the point (5,5) is:

1) $2\sqrt{2}$ **2)** 5 **3)** $4\sqrt{2}$ **4)** 4

19. Let the line $2x + 3y - k = 0$, $k > 0$, intersect the x-axis and y-axis at the point A and B, respectively. If the equation of the circle having the line segment AB as a diameter is $x^2+y^2-3x-2y=0$ and the length of the latus rectum of the ellipse $x^2+9y^2=k^2$ is $\frac{m}{n}$, where 'm' and 'n' are coprime, then $2m + n$ is equal to :

1) 10 **2)** 11 **3)** 13 **4)** 12

20. Match the following Column – I with Column – II :

Column – I	**Column – II**
A) The distance between the line $\vec{r}=\left(2\hat{i}-2\hat{j}+3\hat{k}\right)+\lambda\left(\hat{i}-\hat{j}+4\hat{k}\right)$ and plane $\vec{r}.\left(\hat{i}+5\hat{j}+\hat{k}\right)=5$	P) $\frac{25}{3\sqrt{14}}$
B) Distance between parallel planes $\vec{r}.\left(2\hat{i}-\hat{j}+3\hat{k}\right)=4$ and $\vec{r}.\left(6\hat{i}-3\hat{j}+9\hat{k}\right)+13=0$ is	Q) $\frac{13}{7}$

C) The distance of a point (2, 5, -3) from the plane $\vec{r}.\left(6\hat{i}-3\hat{j}+2\hat{k}\right)=4$ R) $\frac{10}{3\sqrt{3}}$

D) The distance of a point (1, 0, -3) from the plane $x-y-z-9=0$ measured parallel to the line $\frac{x-2}{2}=\frac{y+2}{3}=\frac{z-6}{-6}$ S) 7

1) A – R; B – P; C – S; D – Q **2)** A – R; B – P; C – Q; D – S

3) A – R; B – Q; C – S; D – P **4)** A – R; B – Q; C – P; D – S

SECTION-II (NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. Let 'C' be the largest circle centred at (2, 0) and inscribed in the ellipse $\frac{x^2}{36}+\frac{y^2}{16}=1$. If $(1, \alpha)$ lies on 'C', then $10\alpha^2$ is equal to ……

22. Suppose $\sum_{r=0}^{2023} r^2\ {}^{2023}C_r = 2023\times\alpha\times 2^{2022}$. Then the value of α is …..

23. The number of 9 digit numbers, that can be formed using all the digits of the number 123412341 so that the even digits occupy only even places, is …..

24. The 4th term of GP is 500 and its common ratio is $\frac{1}{m}$, $m\in N$. Let S_n denote the sum of the first 'n' terms of this GP. If $S_6 > S_5 + 1$ and $S_7 < S_6 + \frac{1}{2}$, then the number of possible values of 'm' is …….

25. Let $S=\left\{\theta\in[0, 2\pi]: \tan(\pi\cos\theta)+\tan(\pi\sin\theta)=0\right\}$. Then $\sum_{\theta\in S}\sin^2\left(\theta+\frac{\pi}{4}\right)$ is equal to

PHYSICS **Max Marks: 100**

SECTION-I (SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

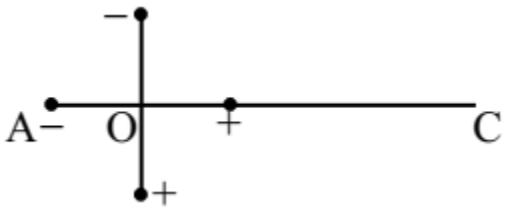
Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

26. If q_f is the free charge on the capacitor plates and q_b is the bound charge on the dielectric slab of dielectric constant 'k' placed between the capacitor plates, then bound charge can be expressed as :

1) $q_b = q_f\left(1-\frac{1}{\sqrt{k}}\right)$ **2)** $q_b = q_f\left(1-\frac{1}{k}\right)$

3) $q_b = q_f\left(1+\frac{1}{\sqrt{k}}\right)$ **4)** $q_b = q_f\left(1+\frac{1}{k}\right)$

27. Two ideal electric dipoles A and B, having their dipole moment p_1 and p_2 respectively are placed on a plane with their centres at O as shown in the figure. At point C on the axis of dipole A, the resultant electric field is making an angle of 37^0 with the axis. The ratio of the dipole moment of A and B, $\frac{p_1}{p_2}$ is : $\left(\text{take } \sin 37^0 = \frac{3}{5}\right)$



1) $\frac{3}{8}$ **2)** $\frac{3}{2}$ **3)** $\frac{2}{3}$ **4)** $\frac{4}{3}$

28. For a gas $C_P - C_V = R$ in a state P and $C_P - C_V = 1.10\ R$ in a state Q, T_P and T_Q are the temperatures in two different states P and Q respectively. Then

1) $T_P = T_Q$ **2)** $T_P < T_Q$ **3)** $T_P = 0.9\ T_Q$ **4)** $T_P > T_Q$

29. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Moment of inertia of a circular disc of mass 'M' and radius 'R' about X, Y axes (passing through its plane) and Z-axis which is perpendicular to its plane were found to be I_x, I_y and I_z respectively. The respective radii of gyration about all the three axes will be the same.

Reason R : A rigid body making rotational motion has fixed mass and shape. In the light of

the above statements, choose the most appropriate answer from the options given below :

1) Both A and R are correct but R is NOT the correct explanation of A.

2) A is not correct but R is correct.

3) A is correct but R is not correct.

4) Both A and R are correct and R is the correct explanation of A

30. A parallel plate capacitor with plate area 'A' and distance of separation 'd' is filled with a dielectric. What is the capacity of the capacitor when permittivity of the dielectric varies as :

$\varepsilon(x) = \varepsilon_0 + kx$, for $\left(0 < x \le \frac{d}{2}\right)$

$\varepsilon(x) = \varepsilon_0 + k(d - x)$, for $\left(\frac{d}{2} \le x \le d\right)$

1) $\left(\varepsilon_0 + \frac{kd}{2}\right)^{2/kA}$ **2)** $\frac{kA}{2\ln\left(\frac{2\varepsilon_0 + kd}{2\varepsilon_0}\right)}$ **3)** 0 **4)** $\frac{kA}{2}\ln\left(\frac{2\varepsilon_0}{2\varepsilon_0 - kd}\right)$

31. In the Young's double slit experiment, the distance between the slits varies in time as $d(t) = d_0 + a_0 \sin\omega t$; where d_0, ω and a_0 are constants. The difference between the largest fringe width and the smallest fringe width obtained over time is given as :

1) $\frac{2\lambda D(d_0)}{(d_0^2 - a_0^2)}$ **2)** $\frac{2\lambda D a_0}{(d_0^2 - a_0^2)}$ **3)** $\frac{\lambda D}{d_0^2} a_0$ **4)** $\frac{\lambda D}{d_0 + a_0}$

32. A linearly polarized electromagnetic wave in vacuum is

$E = 3.1\cos\left[(1.8)z - (5.4\times10^6)t\right]\hat{i}$ N / C

is incident normally on a perfectly reflecting wall at z = a. Choose the correct option

1) The wavelength is 5.4 m

2) The frequency of electromagnetic wave is 54×10^4 Hz

3) The transmitted wave will be $3.1\cos\left[(1.8)z - (5.4\times10^6)t\right]\hat{i}$ N / C

4) The reflected wave will be $3.1\cos\left[(1.8)z + (5.4\times10^6)t\right]\hat{i}$ N / C

33. An electron and proton are separated by a large distance. The electron starts approaching the proton with energy 3 eV. The proton captures the electrons and forms a hydrogen atom in second excited state. The resulting photon is incident on a photosensitive metal of threshold

wavelength 4000 $\overset{0}{A}$. What is the maximum kinetic energy of the emitted photoelectron?

1) 7.61 eV **2)** 1.41 eV

3) 3.3 eV **4)** No photoelectron would be emitted

34. One mole of an ideal gas is taken through an adiabatic process where the temperature rises from 27^0C to 37^0C. If the ideal gas is composed of polyatomic molecule that has 4 vibrational modes, which of the following is true? [R = 8.314 J mol^{-1} k^{-1}]

1) work done by the gas is close to 332 J

2) work done on the gas is close to 582 J

3) work done by the gas is close to 582 J

4) work done on the gas is close to 332 J

35. Match the following Column – I with Column – II :

Column – I	**Column – II**
a) Capacitance, C	i) $M^1L^1T^{-3}A^{-1}$
b) Permittivity of free space, ε_0	ii) $M^{-1}L^{-3}T^4A^2$
c) Permeability of free space, μ_0	iii) $M^{-1}L^{-2}T^4A^2$
d) Electric field, E	iv) $M^1L^1T^{-2}A^{-2}$

Choose the correct answer from the options given below

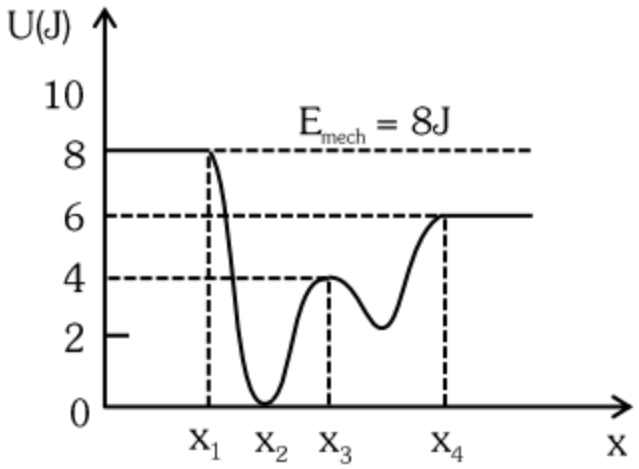
1) (a) $\rightarrow$ (iii), (b) $\rightarrow$ (ii), (c) $\rightarrow$ (iv), (d) $\rightarrow$ (i)

2) (a) $\rightarrow$ (iii), (b) $\rightarrow$ (iv), (c) $\rightarrow$ (ii), (d) $\rightarrow$ (i)

3) (a) $\rightarrow$ (iv), (b) $\rightarrow$ (ii), (c) $\rightarrow$ (iii), (d) $\rightarrow$ (i)

4) (a) $\rightarrow$ (iv), (b) $\rightarrow$ (iii), (c) $\rightarrow$ (ii), (d) $\rightarrow$ (i)

36. Given below is the plot of a potential energy function U(x) for a system, in which a particle is in one dimensional motion, while a conservative force F(x) acts on it. Suppose that $E_{mech} = 8$ J, the incorrect statement for this system is :



[where K.E. = kinetic energy]

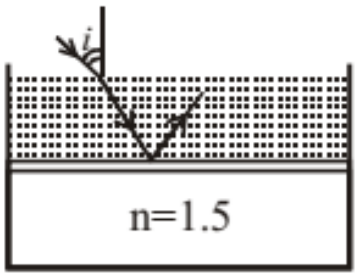
1) at $x > x_4$, K.E. is constant throughout the region.

2) at $x < x_1$, K.E. is smallest and the particle is moving at the slowest speed.

3) at $x = x_2$, K.E. is greatest and the particle is moving at the fastest speed.

4) at $x = x_3$, K.E. = 4 J

37. Consider a tank made of glass(refractive index 1.5) with a thick bottom. It is filled with a liquid of refractive index μ. A student finds that, irrespective of what the incident angle i (see figure) is for a beam of light entering the liquid, the light reflected from the liquid glass interface is never completely polarized. For this to happen, the minimum value of μ is :



1) $\frac{3}{\sqrt{5}}$ **2)** $\frac{5}{\sqrt{3}}$ **3)** $\sqrt{\frac{5}{3}}$ **4)** $\frac{4}{3}$

Consider the following statements :

A) Atoms of each element emit characteristics spectrum.

B) According to Bohr's Postulate, an electron in a hydrogen atom, revolves in a certain stationary orbit.

C) The density of nuclear matter depends on the size of the nucleus.

D) A free neutron is stable but a free proton decay is possible.

E) Radioactivity is an indication of the instability of nuclei.

Choose the correct answer from the options given below :

1) A, B, C, D and E **2)** A, B and E only

3) B and D only **4)** A, C and E only

39. A particle of mass M originally at rest is subjected to a force whose direction is constant but magnitude varies with time according to the relation

$$F = F_0\left[1-\left(\frac{t-T}{T}\right)^2\right]$$

Where F_0 and T are constants. The force acts only for the time interval 2T. The velocity v of the particle after time 2T is :

1) $2F_0T/M$ **2)** $F_0T/2M$ **3)** $4F_0T/3M$ **4)** $F_0T/3M$

40. Given below are two statements :

Statement-I : The reactance of an ac circuit is zero. It is possible that the circuit contains a capacitor and an inductor.

Statement-II : In ac circuit, the wattless component of current delivered by the source never becomes zero.

In the light of the above statements, choose the

correct answer from the options given below :

1) Both Statement I and Statement II are true.

2) Both Statement I and Statement II are false.

3) Statement I is true but Statement II in false.

4) Statement I is false but Statement II is true

41. Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : In an uniform magnetic field, speed and energy remains the same for a moving charged particle.

Reason (R) : Moving charged particle experiences magnetic force perpendicular to its direction of motion.

1) Both (A) and (R) are true and (R) is the correct explanation of (A)

2) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

3) (A) is true but (R) is false

4) (A) is false but (R) is true

42. Two identical cells each of emf 1.5 V are connected in parallel across a parallel combination of two resistors each of resistance 20Ω. A voltmeter connected in the circuit measures 1.2 V. The internal resistance of each cell is

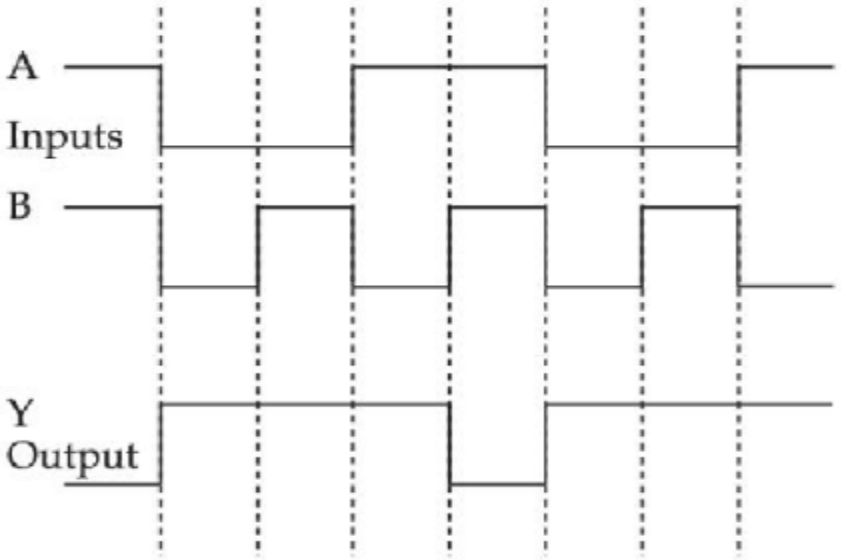
1) 2.5 Ω **2)** 4 Ω **3)** 5 Ω **4)** 10 Ω

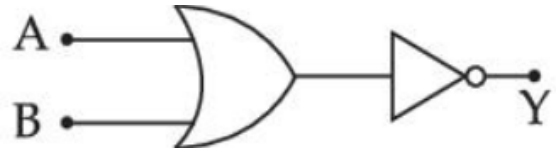
43. A plane electromagnetic wave travels in a medium of relative permeability 1.61 and relative permittivity 6.44. If magnitude of magnetic intensity is 4.5×10^{-2} Am^{-1} at a point, what will be the approximate magnitude of electric field intensity at that point ?

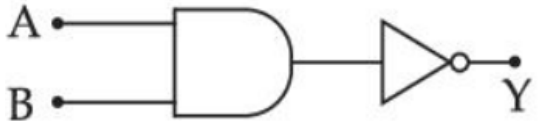
(Given : permeability of free space $\mu_0 = 4\pi\times10^{-7}$ S.I units, speed of light in vacuum $c = 3\times10^{8}\ ms^{-1}$)

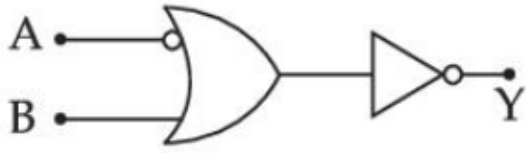
1) 16.96 Vm^{-1} **2)** 2.25×10^{-2} Vm^{-1} **3)** 4.2×10^{6} Vm^{-1} **4)** 6.75×10^{6} Vm^{-1}

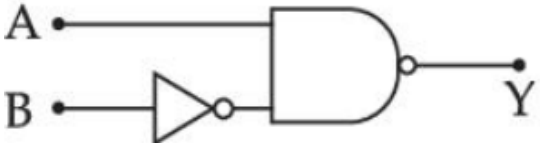
44. Identify the correct Logic Gate for the following output (Y) of two inputs A and B.



1)


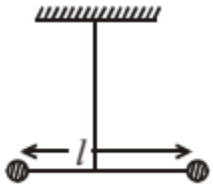
2)


3)


4)


45. Two masses m and $\frac{m}{2}$ are connected at the two ends of a massless rigid rod of length *l*. The rod is suspended by a thin wire of torsional constant k at the centre of mass of the rod-mass system(see figure). Because of torsional constant k, the restoring torque is $\tau = k\theta$ for

angular displacement 0. If the rod is rotated by θ_0 and released, the tension in it when it passes through its mean position will be:



1) $\frac{3k\theta_0^2}{l}$ **2)** $\frac{k\theta_0^2}{2l}$ **3)** $\frac{2k\theta_0^2}{l}$ **4)** $\frac{k\theta_0^2}{l}$

SECTION-II (NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

46. From the given data, the amount of energy required to break the nucleus of aluminium $^{27}_{13}Al$ is …….. $x\times10^{-3}$ J

Mass of neutron = 1.00866 u

Mass of proton = 1.00726 u

Mass of Aluminium nucleus = 27.18846 u

(Assume 1 u corresponds to x J of energy)

(Round off to the nearest integer)

47. A circular conducting coil of radius 1 m is being heated by the change of magnetic field $\vec{B}$ passing perpendicular to the plane in which the coil is laid. The resistance of the coil is $2\mu\Omega$. The magnetic field is slowly switched off such that its magnitude changes in time as

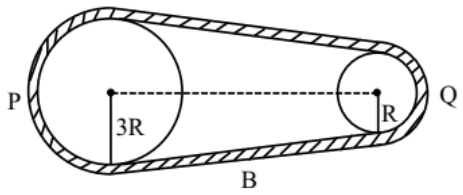
$$B=\frac{4}{\pi}\times10^{-3}T\left(1-\frac{t}{100}\right)$$

The energy dissipated by the coil before the magnetic field is switched off completely is E = …… mJ.

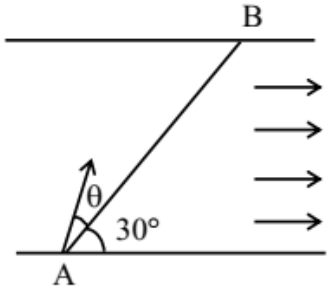
48. The value of aluminium susceptibility is 2.2×10^{-5} . The percentage increase in the magnetic field if space within a current carrying toroid is filled with aluminium is $\frac{x}{10^4}$. Then the value of 'x' is ……..

49. In the given figure, two wheels P and Q are connected by a belt B. The radius of P is three times as that of Q. In case of same rotational kinetic energy, the ratio of rotational inertias $\left(\frac{I_1}{I_2}\right)$ will be x : 1. The value of x will be …….

I_1 is rotational inertia of P, I_2 is rotational inertia of Q



50. A swimmer wants to cross a river from point A to point B. Line AB makes an angle of 30^0 with the flow of river. Magnitude of velocity of the swimmer relative to water is same as that of the river. The angle θ with the line AB that the swimmer should swim should be …… (in degrees), so that the swimmer reaches point B.



CHEMISTRY

Max Marks: 100

SECTION-I (SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

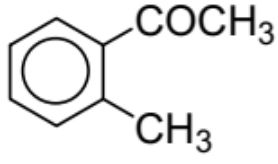
51. An aromatic compound 'X' with molecular formula $C_9H_{10}O$ gives the following chemical tests:

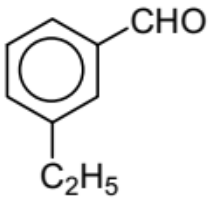
I) forms 2, 4-DNP derivative

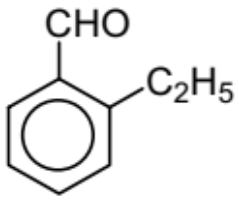
II) reduces Tollens' reagent

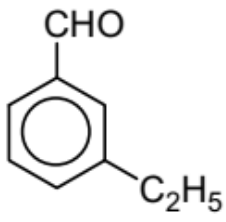
III) undergoes Cannizzaro reaction and

IV) on vigorous oxidation, 1, 2-benzenedicarboxylic acid is obtained, X is

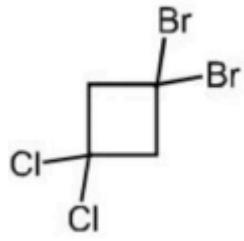
1) 

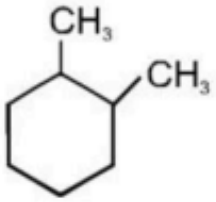
2) 

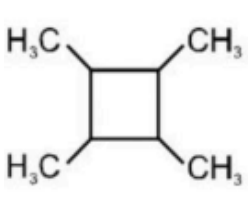
3) 

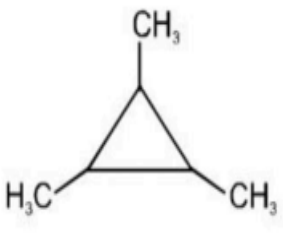
4) 

52. Which of the following compound cannot show geometrical isomerism?

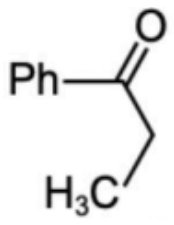
1) 

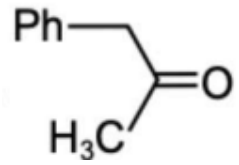
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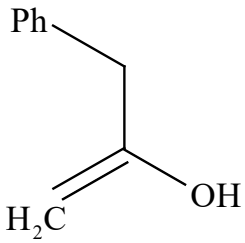
3) 

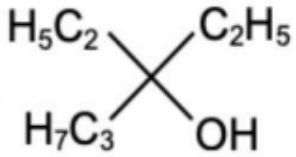
4) 

53. In the reaction $Ph-C\equiv C-CH_3 \xrightarrow{Hg^{2+}/H^+} A$, A is

1) 

2) 

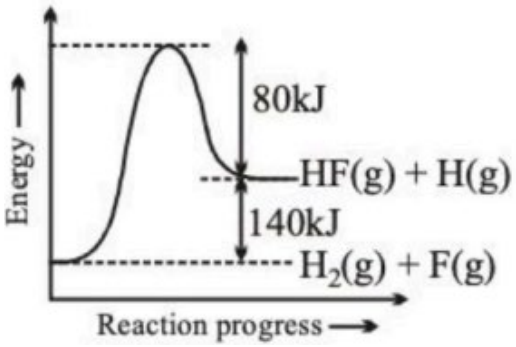
3) 

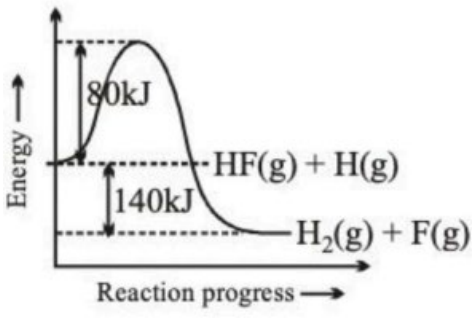
4) 

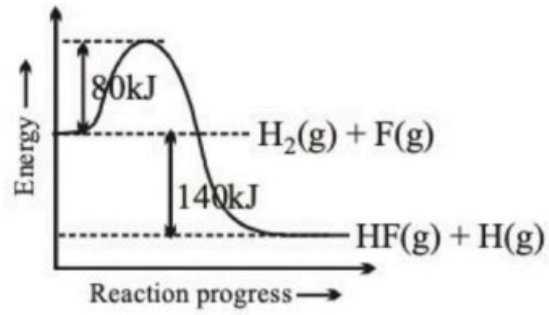
54. For electrometry reaction :

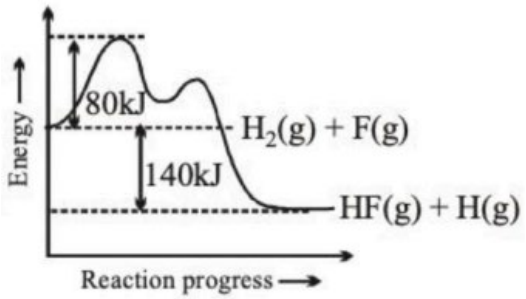
$H_2(g)+F(g)\rightarrow HF(g)+H(g) \qquad \Delta H=-140\ kJ/mol$

Activation energy is 80 kJ/mol. Then correct energy diagram is :

1) 

2) 

3) 

4) 

55. Column – I contains four entries and Column – II contain four entries. Entry of Column – I are to be matched with only one entry of Column – II

Column – I	**Column – II**
A) Phenol + Aniline	P) +ve deviation from Raoult's law
B) Ethanol + Acetone	Q) Ideal solution
C) n-hexane + n-heptane	R) Colligative property
D) Osmotic pressure	S) –ve deviation from Raoult's law

1) A – S; B – P; C – Q; D – R

2) A – R; B – S; C – Q; D – P

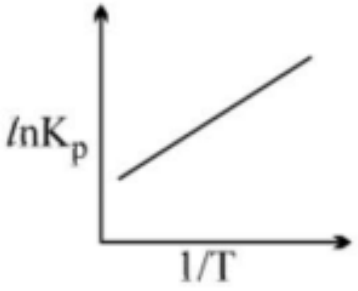
3) A – Q; B – R; C – S; D – P

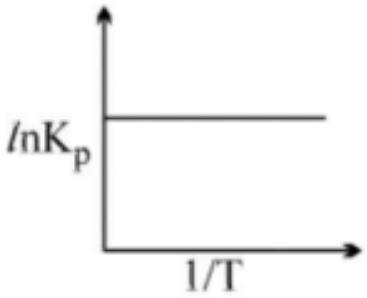
4) A – P; B – S; C – Q; D – R

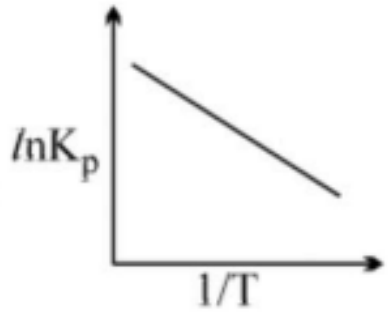
56. Find true and False from the following statements regarding carbohydrates
Statement – I : Bromine water can be used to differentiate between aldoses and ketoses
Statement – II : A pair of diastereomeric aldoses which differ only in configuration at C-2 are anomers.
1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
3) Statement-1 is True, Statement-2 is False
4) Statement-1 is False, Statement-2 is True

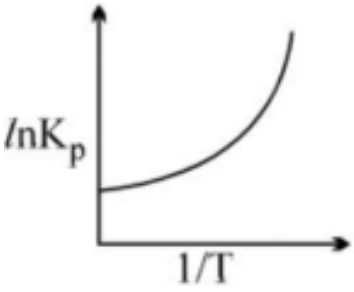
57. An amine is treated with HNO_2 and subsequently further treated with few drops of phenol and con H_2SO_4. A blue green colour is developed on adding water it changes to red by adding NaOH, it further changes to dark blue the amine is.
1) Aniline **2)** N-methyl aniline
3) O-methyl aniline **4)** N, N-methyl aniline

58. An exothermic reaction is represented by the graph :

1)


2)


3)


4)


59. Assertion (A) : Higher the R_f value of a substance, greater is the time taken by it for elution.
Reason (R) : The time taken by a substance for elution is greater when it is adsorbed more strongly on the absorbent.
1) Both (A) and (R) are true and (R) is the correct explanation of (A)
2) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)
3) (A) is correct but (R) is incorrect **4)** (A) is incorrect but (R) is correct

60. Which of the following element shows maximum valency?
1) Carbon **2)** Barium **3)** Nitrogen **4)** Sulphur

61. Find the correct order of electron gain enthalpy of the given elements:(consider magnitude only)

1) O > B > C > N **2)** O > C > N > B **3)** O > C > B > N **4)** O > N > C > B

62. Which compound is called as inorganic benzene?

1) $B_3N_3H_6$ **2)** $B_3N_3H_3$ **3)** BN_3H_3 **4)** $B_6N_3H_3$

63. For the following Assertion (A) and Reason (R), the correct option is :

Assertion (A) : The pH of water increases with increase in temperature

Reason (R) : The dissociation of water into H^+ and OH^- is an exothermic reaction.

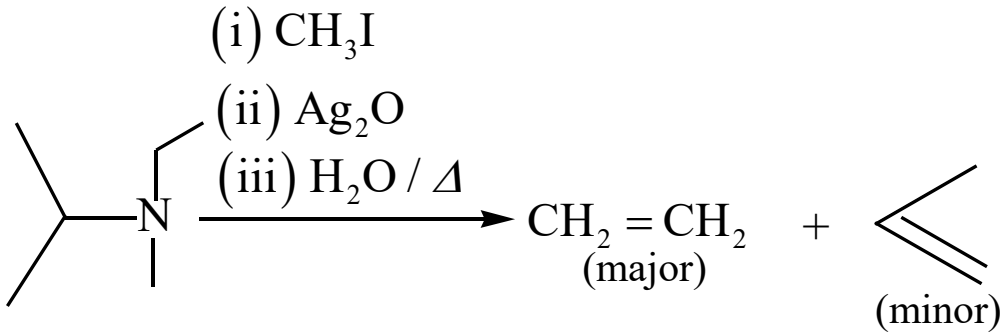
1) (A) is false but (R) is true

2) Both (A) and (R) are true and (R) is the correct explanation of (A)

3) Both (A) and (R) are false

4) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

64. Statement – I:



Statement – II : It follows Hofmann's elimination by E_2

1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1

3) Statement-1 is True, Statement-2 is False

4) Statement-1 is False, Statement-2 is True

65. Pressure of 10 moles of an ideal gas is changed from 2atm to 1atm against constant external pressure without change in temperature. If surrounding temperature (300 K) and pressure (1atm) always remains constant then calculate total entropy change $\left(\Delta S_{system} + \Delta S_{surrounding}\right)$ for given process.

[Given: ln2 = 0.70 and R = 8.0 J/mol/K]

1) 56 J/K **2)** 14 J/K **3)** 16 J/K **4)** None of these

66. What weight of glucose must be dissolved in 100 g of water to lower the vapour pressure by 0.50 mm Hg?

(Assume dilute solution is being formed)

Given: Vapour pressure of pure water is 65.5 mm Hg at room temperature. Molar mass of glucose is 180g mol^{-1}.

1) 4.69 g **2)** 3.69 g **3)** 2.59 g **4)** 7.63 g

67. When a hydrocarbon A undergoes combustion in the presence of air, it required 7.5 equivalents of oxygen and produces 3 equivalents of water. What is the molecular formula of A?

1) C_8H_6 **2)** C_9H_9 **3)** C_6H_6 **4)** C_9H_6

68. Radius of the first excited state of Helium ion is given as :

$a_0 \rightarrow$ radius of first stationary state of hydrogen atom.

1) $r = \frac{a_0}{2}$ **2)** $r = \frac{a_0}{4}$ **3)** $r = 4a_0$ **4)** $r = 2a_0$

69. The primary and secondary valencies of copper respectively in $\left[Cu\left(NH_3\right)_4\right]Cl_2$ are :

1) 3 and 5 **2)** 2 and 6 **3)** 2 and 8 **4)** 2 and 4

70. A solution of aluminium chloride is electrolysed for 30 minutes using a current of 2A. The amount of the aluminium deposited at the cathode is ……

[Given : molar mass of aluminium and chloride are 27 $g\,mol^{-1}$ and 35.5 g mol^{-1} respectively, Faraday constant = 96500 $C\,mol^{-1}$]

1) 1.660 g **2)** 1.007 g **3)** 0.336 g **4)** 0.441 g

SECTION-II (NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

71. If the CFSE of $\left[Ti\left(H_2O\right)_6\right]^{3+}$ is -192 kJ / mol, this complex will absorb maximum at wavelength …….. nm. (nearest integer)

Assume Planck's constant $(h) = 6.4 \times 10^{-34}$ Js speed of light (c) = 3.0×10^{8} m / s and Avogadro's constant $(N_A) = 6 \times 10^{23}$ / mol.

72. The rate constant for a first order reaction is $20\ \text{min}^{-1}$. The time required for the initial concentration of the reactant to reduce to its $\frac{1}{64}$ level is $\ldots\ldots \times 10^{-2}$ min. (Nearest integer)

73. The metal M can exist in two oxidation states + 2 and +3. In the sample of $M_{0.83}\ O_{1.00}$, the percentage of metal ions existing in +2 oxidation state is ….. % (nearest integer)

74. At 298 K

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g),\ K_1 = 8 \times 10^5$

$N_2(g) + O_2(g) \rightleftharpoons 2NO(g),\ K_2 = 1.6 \times 10^{12}$

$H_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons H_2O(g),\ K_3 = 2 \times 10^{-13}$

Based on above equilibria, the equilibrium constant of the reaction,

$2NH_3(g) + \frac{5}{2}O_2(g) \rightleftharpoons 2NO(g) + 3H_2O(g)$ is $\ldots\ldots \times 10^{-33}$ (nearest integer)

75. How many geometrical isomers of major product (A) are possible?

